Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 13

B.Pharma (Sem.-4)

PHYSICAL PHARMACEUTICS-II

Subject Code : BP-403T

M.Code : 75845

Date of Examination : 04-01-2023

Time : 3 Hrs. **Max. Marks : 75**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. Answer briefly :

a) Pseudoplastic flow *vs.* Dilatant flow.

b) Number distribution *vs.* weight distribution curve.

c) Any two advantages of cone plate viscometer over cup bob viscometer.

d) Dielectric constant.

e) Difference between flocculated and deflocculated suspensions.

f) Kinematic viscosity.

g) Elastic modulus.

h) Enumerate different theories of emulsification.

i) Sieving.

j) Acid base catalysis.

SECTION-B

2. Discuss Principle, construction, working and applications of Brookefield viscometer.

3. State different methods of determining Particle size and discuss Andreason Pipette method in detail.

4. Discuss stability of emulsions.

SECTION-C

5. A drug solution of 150 mg/ml concentration was prepared. Calculate the time required for 20% decomposition of drug, if decomposition follows a zero order kinetics for which specific reaction rate $K = 2.5 \times 10^{-2}$ moles/Lt/hr

6. Define zeta potential. How it is determined?

7. Define Thixotropy. Explain its importance in pharmacy.

8. Explain Arrhenius theory of electrolytic dissociation.

9. Enumerate the important factors that influence the particle size analysis by sieving.

10. Discuss the stability of pharmaceutical products against hydrolysis.

11. Discuss plastic *vs* elastic deformation in solids

12. Discuss the optical *vs* electrical properties of colloidal dispersions.

13. Discuss in brief the interfacial properties of suspensions.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.